

Ashwin Conrad

219 Parkside Cress, SE
Calgary, AB, T2J4J3
(403) 826-9320
conradashwin@gmail.com
ashwinconrad.ca

I am dedicated to continuous learning and excited to contribute innovative solutions to a company that shares my values of sustainability, responsibility, and respect.

EDUCATION

University of Alberta, Edmonton, AB - Mechanical Engineering - 2023 - Present

- **2nd Year Engineering Student**
- Extra Curricular: Engineering Student Engagement Leader, Society of Automotive Engineers
Formula Racing Club Member

West Island College, Calgary, AB - Highschool Diploma - 2020 - 2023

- Extra Curricular: LADS Clubs (Male Leadership in the Local Community), Diversity Inclusion
Pluralism Committee, Senior Varsity Basketball and Rugby

TECHNICAL SKILLS

- Office 365, AutoCAD & 2D laser cutting, Excel data and statistical analysis, LoggerPro,
3D-modeling & printing, Python, Java

EXPERIENCE

Formula EV Racing, University of Calgary - Low Voltage Electronics Lead - Winter 2025 - Present

- Lead low-voltage electrical system planning and integration for Formula UA27, coordinating
board-level hardware, safety interfaces, and vehicle packaging with the broader team.

AltaGas, Calgary, AB - Engineering Co-op - 2025 - 2026

- Built automation and data-analysis workflows for corrosion monitoring, inspection planning, and
asset lifecycle work across multi-facility operations.

Spartan Controls, Calgary, AB - Control Panel Technician - Summer Terms, '23, '24

- Gained hands-on experience in remote transmission units, RTU, and electrical departments,
focusing on panel production and quality assurance.
- Completed 15+ multi-panel projects in individual, partner-based, and group setups, consistently
meeting and exceeding expectations.
- Interpreted and analyzed engineering drawings, panel designs, and electrical schematics.
- Worked extensively with advanced panel systems, including various industry-standard Distributed
Control Systems, DCS, and signal transmission and power supply units.
- Resolved quality control issues accurately and met strict deadlines.
- Assisted with pressure and instrumentation testing for RemVue Control Panels and processed
250+ PLC software updates in under half the allocated time.
- Collaborated in team environments emphasizing safety and efficiency.
- Applied technical understanding and precision skills in panel assembly while adhering to high
standards and identifying deficiencies.

Cherry Pit, Calgary, AB - Floor Worker/Cashier - Mar 2022 - Jul 2023

- Stocked vegetable displays and provided customer service.
- Conducted timely deliveries to customers and businesses.
- Managed and organized incoming pallets.
- Assisted in training new hires.

A-Plus Parking, Calgary, AB - Service Technician - Dec 2019 - Sep 2022

- Completed general maintenance and troubleshooting repairs on trucks, trailers, and old/broken
equipment.
- Silicon moulded and 2 part casted bumper plates for street-sweeping equipment, reducing the
cost of single-use products by 50%.
- Assisted with bookkeeping, painting parkade lines, and cleaning.

SKILLS

Leadership & Teamwork -
Organization - Trouble
Shooting/Problem Solving -
Fast Learner

AWARDS

Undergraduate Student
Leadership Award Winner -
2024-2025

Nomination for Best Project -
Eng 160 Final Project 2024

Leadership in Engineering
Scholarship Winner 2023-2024

APEGA Science Olympics
Medalist: Silver 2021-2022,
Silver 2022-2023

VOLUNTEERING

Engineering Student
Engagement Leader
2024-2025

CYDC Assistant Youth
Basketball Coach 2024

WIC Grade 7-8 Head &
Assistant Basketball Coach
2022-2023

PERSONAL INTERESTS

Cooking - Physical Fitness -
metal/wood/leather/composite
material design and projects

Project & Club Portfolio

ENGINEERING PROJECTS

AltaGas, Microsoft Fabric, Power BI, Python - Corrosion Monitoring Dashboard - 2025 - 2026

- Built a centralized data workflow for corrosion coupon tracking across 12 facilities and 100+ monitored assets.
- Reduced manual vendor-report processing by structuring incoming data for repeatable Power BI reporting and analysis.

AltaGas, Python, openpyxl, python-docx - Inspection Scope Generation Tool - 2025 - 2026

- Developed a Python tool to generate 500+ equipment-specific inspection scopes from database exports.
- Reduced manual document creation time by standardizing Excel-to-Word scope generation.

AltaGas, Python, data analysis, technical assessment - Heat Exchanger Lifecycle Analysis - 2025 - 2026

- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers to assess fatigue risk and monitoring gaps.
- Summarized lifecycle findings into practical recommendations for inspection planning and operating awareness.

Spartan Controls, Industrial controls and panel QA - Control Panel Builds - *Summer Terms, '23, '24*

- Built, wired, loaded, and checked a sample of 30+ electrical control panels across individual and team builds.
- Used schematics, wiring standards, and pre-delivery quality checks to resolve deficiencies before customer delivery.

Spartan Controls, Shop workflow tooling - Leather Tool Carrier - 2024

- Designed and built a leather tool carrier to keep common hand tools organized and accessible during panel assembly.
- Turned a personal workflow problem into a durable shop aid for faster repeated work.

CLUBS & TEAMS

Formula EV Racing, Low-voltage electrical systems - Formula EV Electrical Systems - *Winter 2025 - Present*

- Coordinate low-voltage power distribution, sensor integration, and safety-interface planning around the UA27 vehicle package.
- Work across mechanical and software constraints to make electrical systems accessible, serviceable, and competition-ready.

FABRICATION & MATERIALS

Personal Project, Lamination, fiberglass layups, embedded aluminum - Composite Skateboard Decks - *Earlier build series*

- Built skateboard decks while learning lamination and fiberglass layup techniques, including one green deck with sealed-in aluminum.

Personal Project, Forging, heat, grinding, finishing - Father's Day Knife Forging - 2024

- Completed a knife forging project focused on shaping, heat response, grinding, and practical finish work.